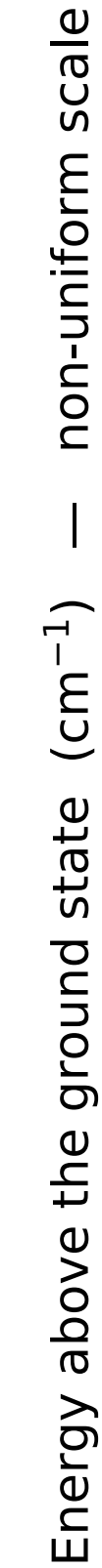
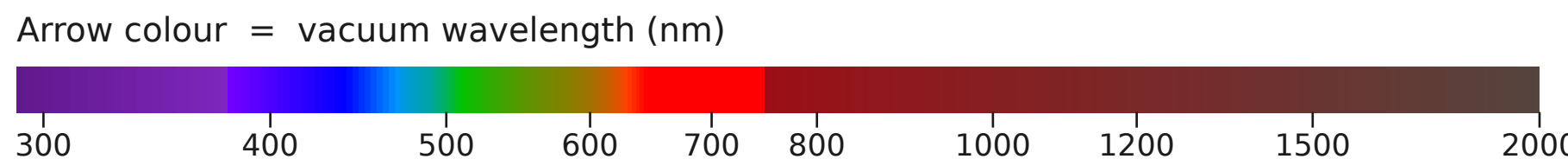


Ne (Ne I). Strongest classified transitions below 2 μm (NIST): 49 levels, 74 lines.

Ne⁺ ionisation limit 173929.75 cm⁻¹ = 21.56454 eV (57.494 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $\langle J || er || J' \rangle$ in ea_0
 Matrix elements are measured or NIST values; * = high-accuracy theory, $\approx$ = model or semi-empirical calculation.
 Arrow width grows with the Einstein coefficient A . Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
 Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists)



Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ \langle J e J' \rangle $ (ea)	A (s^{-1})	$\Gamma/2\pi$ up. (MHz)	branch (%)	Use
$^1S_0 - 2(1/2)^1_1$	73.5896	-	4073.8413	0.589	5.880e+08	97.16	96.3	73.6 nm resonance line
$^1S_0 - 2(3/2)^1_1$	74.3720	-	4030.9880	0.164	4.400e+07	7.978	87.8	74.4 nm resonance line
$^1S_0 - 2[3/2]^1_2$	74.6036	-	4018.4733	-	6.790e-02	1.08e-08	100	74.6 nm M2 decay of the metastable 3P2 state
$2(3/2)^1_2 - 2(3/2)^1_2$	614.4763	614.3063	487.8829	4.018	2.820e+07	8.079	55.5	614.3 nm line used for collinear laser spectroscopy of Ne...
$2(3/2)^1_2 - 2[5/2]_1$	640.4017	640.2247	468.1319	6.835	5.149e+07	8.162	100	640.2 nm laser-cooling / MOT line

λ , ν from NIST level energies unless a measured frequency for this isotope is available. A : measured value or NIST. $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.

²¹Ne hyperfine structure ($I = 3/2$)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
$2[3/2]_2^o$	-267.68(3)	-111.55(10)	F=1/2: +1106.95 F=3/2: +803.04 F=5/2: +203.56 F=7/2: -830.93
$2[5/2]_2^o$	-142.40(20)	-107.71(11)	F=3/2: +789.78 F=5/2: +514.55 F=7/2: +53.85 F=9/2: -667.73
$2[3/2]_1^o$	-185.83(21)	-	F=1/2: +836.24 F=3/2: +557.49 F=5/2: +92.92 F=7/2: -557.49

Measured A, B constants; sources in the data file.

Isotope shifts			
Transition	λ vac (nm)	Isotopes	Shift (MHz)
$1^1_0 - 2 3/2\rangle^1$	62.974	22-20	2298(31)
$1^1_0 - 2 1/2\rangle^1$	73.590	22-20	(90-90)
$1^1_0 - 2 3/2\rangle^1$	74.372	22-20	420(90)
$2 3/2\rangle^1 - 2 1/2\rangle_0$	540.206	22-20	1804(26)
$2 1/2\rangle^1 - 2 1/2\rangle_0$	585.411	22-20	2286.3(27)
$2 3/2\rangle^1 - 2 1/2\rangle_1$	588.352	22-20	1737.0(5)
$2 3/2\rangle^1 - 2 3/2\rangle_1$	594.648	22-20	1716.0(11)
$2 3/2\rangle^1 - 2 3/2\rangle_1$	597.719	22-20	1727.4(19)
$2 3/2\rangle^1 - 2 1/2\rangle_1$	603.167	22-20	1744(5)
$2 3/2\rangle^1 - 2 1/2\rangle_0$	607.602	22-20	1702.0(25)
$2 3/2\rangle^1 - 2 3/2\rangle_1$	609.785	22-20	1725.3(25)
$2 3/2\rangle^1 - 2 3/2\rangle_1$	613.015	22-20	1737.6(20)
$2 3/2\rangle^1 - 2 3/2\rangle_1$	614.476	22-20	1663.0(12)
$2 3/2\rangle^1 - 2 3/2\rangle_1$	614.476	21-20	874.94
$2 1/2\rangle^1 - 2 1/2\rangle_1$	616.530	22-20	1656.4(28)
$2 1/2\rangle^1 - 2 3/2\rangle_1$	621.900	22-20	1669.43(8)
$2 1/2\rangle^1 - 2 3/2\rangle_1$	626.823	22-20	1647.38(14)
$2 3/2\rangle^1 - 2 3/2\rangle_1$	630.653	22-20	1674.27(19)
$2 3/2\rangle^1 - 2 5/2\rangle_1$	633.618	22-20	1641.05(16)
$2 3/2\rangle^1 - 2 3/2\rangle_1$	638.476	22-20	1679.60(26)
$2 3/2\rangle^1 - 2 5/2\rangle_1$	640.402	22-20	1626.05(11)
$2 3/2\rangle^1 - 2 5/2\rangle_1$	640.402	21-20	855.7(10)

$$\text{Shift} = \nu(\text{first isotope}) - \nu(\text{second isotope})$$